Donald Shenaj

Postdoctoral Researcher

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Research Interests

Model Merging, Model Patching, Image Generation, LLMs, Federated Learning, Continual Learning, Domain Adaptation.

Work Experience

May 2025 - Now **Postdoctoral Researcher**, University of Pisa, Italy.
CoEvolution (EU Horizon Europe) – Adaptive AI for Evolving Cybersecurity Threats.

Oct. 2024 - Apr. 2025 **Research Intern**, Samsung Research, UK.
Supervisors: Umberto Michieli, Mete Ozay.

May 2023 - Nov. 2023 **Visiting Researcher**, Mila - Quebec AI Institute & Concordia University, Canada.
Supervisor: Eugene Belilovsky.

Oct. 2021 - Sep. 2024 **Teaching Assistant**, University of Padova, Italy.
M.Sc. courses: Computer Vision (22/23, 23/24), Scientific computing with python (22/23), Machine Learning (21/22, 23/24).

Education

Oct. 2021 - Mar. 2025 **Ph.D. in Information Engineering**, University of Padova, Italy.
Thesis: Morphing Distributed and Transfer Learning Paradigms for Visual Understanding.
Successfully defended on 2025/03/24.
Supervisor: Pietro Zanuttigh.
Seasonal schools: IEEE-EURASIP S3P 2024, GTTI 2024, Mila GFlowNet 2023, CIFAR DL+RL 2023, GTTI 2023, IEEE-DEI SSIE 2022, GTTI 2022.
Co-Advised: More than 10 Master's and Bachelor's theses.

Oct. 2019 - Sep. 2021 **M.Sc. in ICT for Internet and Multimedia**, University of Padova, Italy.
Grade: 110/110 cum Laude.
Thesis: Coarse-to-Fine Learning for Semantic Segmentation across Multiple Domains.
Supervisor: Pietro Zanuttigh.

Sep. 2016 - Oct. 2019 **B.Sc. in Electronics Engineering**, University of Bologna, Italy.
Grade: 110/110 cum Laude.
Thesis: Implementation and analysis of a vehicle counter system with Python and OpenCV.
Supervisor: Enrico Paolini.

Skills

- Programming: C, C++, Python, Matlab/GNU Octave, CUDA, Bash, VHDL, LabVIEW, Java, JavaScript
- Typesetting: LaTeX, Manim, Markdown, HTML, CSS
- Libraries: PyTorch, TensorFlow, Keras, OpenCV, Numpy, SciPy, Pandas, Matplotlib, Scikit-learn
- System: Linux, Git, HPC clusters, Slurm, Singularity, Docker
- Languages: Italian (Native), English (Professional), Albanian (Elementary), French (Elementary)

Interests

Programming, Rubik's Cubes, Music, Movies/TV Series, Skateboarding, Street art

Reviewer

ICCV 2025, CVPR 2025 (Outstanding Reviewer, top 5% of 12,593 reviewers), AAAI 2025, ICLR Workshops 2025, CVPR Workshops 2025, WACV 2025, IEEE TPAMI, IEEE TMM, Pattern Recognition, CVIU, MMSP 2024, ICML 2023 Workshops, ICPR 2022, Harms and Risks of AI in the Military (HRAIM)

Publications

Preprints

[P1] **D. Shenaj**, O. Bohdal, T. Ceritli, M. Ozay, P. Zanuttigh, U. Michieli, "K-Merge: Online Continual Merging of Adapters for On-device Large Language Models" (2025, under review).

[P2] M. Caligiuri, F. Barbato, **D. Shenaj**, U. Michieli, P. Zanuttigh, "FedPromo: Federated Lightweight Proxy Models at the Edge Bring New Domains to Foundation Models" (2025, under review).

Conferences

[C1] **D. Shenaj**, O. Bohdal, M. Ozay, P. Zanuttigh, U. Michieli, "LoRA.rar: Learning to Merge LoRAs via Hypernetworks for Subject-Style Conditioned Image Generation", IEEE/CVF International Conference on Computer Vision (**ICCV**), 2025.

[C2] **D. Shenaj**, E. Belilovsky, P. Zanuttigh, "Adaptive Local Training in Federated Learning", **ICLR** 2025 Workshop on Modularity for Collaborative, Decentralized, and Continual Deep Learning and **EUSIPCO** 2025.

[C3] G. Rizzoli*, M. Caligiuri*, **D. Shenaj**, F. Barbato, P. Zanuttigh, "When Cars meet Drones: Hyperbolic Federated Learning for Source-Free Domain Adaptation in Adverse Weather", IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**) [acceptance rate first round=12.1%] (Oral), 2025.

[C4] G. Rizzoli, **D. Shenaj**, P. Zanuttigh, "Source-Free Domain Adaptation for RGB-D Semantic Segmentation with Vision Transformers", IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), Pretrain Workshop (Oral), 2024.

[C5] **D. Shenaj**, M. Toldo, A. Rigon, P. Zanuttigh, "Asynchronous Federated Continual Learning", IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), FedVision Workshop (Oral), 2023.

[C6] **D. Shenaj***, E. Fani*, M. Toldo, D. Caldarola, A. Tavera, U. Michieli†, M. Ciccone†, P. Zanuttigh†, B. Caputo†, "Learning Across Domains and Devices: Style-Driven Source-Free Domain Adaptation in Clustered Federated Learning", IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**) [acceptance rate first round=22.3%] (Oral), 2023.

Journals

[J1] **D. Shenaj***, G. Rizzoli*, P. Zanuttigh, "Federated Learning in Computer Vision", **IEEE Access**, 2023.

[J2] **D. Shenaj**, F. Barbato, U. Michieli, P. Zanuttigh, "Continual coarse-to-fine domain adaptation in semantic segmentation", Image and Vision Computing (**IMAVIS**), 2022.

* indicates equal contribution, † indicates equal supervision

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